Topic Modeling of the Social Anxiety Subreddit:

This project applies three topic modeling approaches: Latent Dirichlet Allocation (LDA), BERTopic, and TopicGPT to 27,197 posts and 238,589 comments from the r/socialanxiety subreddit collected between March 1, 2019 and March 1, 2021. This overview provides a brief introduction to topic modeling in the context of social media text analysis, followed by descriptions of data preprocessing and model implementation. Results of each modeling approach are then presented along with a discussion of their relative strengths, limitations, and overall utility for social science research.

Overall, LDA and BERTopic proved effective for identifying broader themes across the dataset, while BERTopic offered faster performance and more intuitive tools for exploring topics. BERTopic also performed well at identifying smaller subtopics when modeling a higher number of clusters. TopicGPT showed potential for generating interpretable topic hierarchies and subtopics, though its performance depended heavily on prompt design and computational resources.

Reddit and pushshift

Reddit is an online forum made up of communities called subreddits. Each subreddit focuses on a specific topic, and users can create posts and leave comments in response to others. The subreddit r/socialanxiety is dedicated to discussions about social anxiety.

Pushshift is an archive of Reddit posts and comments that enables large-scale analysis of past discussions. The collected data is stored in JSONL format, where each line contains a set of key-value pairs including information such as the creation date and author of the post or comment. Two separate JSONL files were collected: one for posts and one for comments. For this project, posts concatenated with their comments from r/socialanxiety between March 1, 2019 and March 1, 2021 were analyzed. This dataset was chosen because it matched a previous publication (Manova et al. 2024) comparing posts before and during the Coronavirus pandemic, which provided a benchmark for assessing the utility of the different methods.

Topic modeling

Topic modeling is the use of computational methods to identify themes within a corpus. A corpus is a collection of documents, and documents are units of text data such as a news article, Wikipedia entry, or Reddit post. These methods help summarize large collections of text by identifying patterns in how words appear across documents. Other applications include analyzing how information spreads across online social networks, gathering features for further machine

learning tasks, discovering health-related topics, and understanding user perceptions on specific subjects (Curiskis et al., 2019).

Latent Dirichlet Allocation (LDA) assumes that each document is made up of a user-defined number of topics and that each topic is defined by a distribution of words. By examining the frequency of words across documents, LDA estimates both the topics present in the corpus and how strongly each document is associated with them. Each topic can then be manually labeled according to its most frequently occurring words (for example: school, class, teacher, high, presentation, college → posts discussing school).

BERTopic is a more recent topic modeling approach that uses transformer-based encoding models to represent documents as vectors (lists of numbers). These vector representations are then reduced in dimensionality and clustered so that similar documents are grouped together into topics. As with LDA, overrepresented words within each cluster can be used to label the resulting topics.

TopicGPT is a prompt-based topic modeling approach that uses a large language model to identify and refine topics directly from documents. Rather than analyzing statistical patterns across the entire corpus, TopicGPT can optionally be given a list of seed topics with descriptions and then reads documents one by one. For each document, it is instructed to categorize it by creating a new topic or assigning the document to an existing topic(s). In this project, TopicGPT was also used to further categorize documents within topics into subtopics, modeling its level of specificity based on curated prompt examples.

Pushshift Data Preprocessing

Posts and comments were first loaded using the Python libraries pandas and json. Both datasets were filtered to include only items within the desired timeframe (March 2019 – March 2021). Posts and comments where the body text was empty, deleted, or removed were excluded from the dataset.

Post bodies were appended to titles, and all comments associated with each post were appended sequentially according to their creation date, separated by spaces. Text was cleaned using regular expressions to remove emoticons, symbols, URLs, HTML formatting, and asterisks used for bullet points or italics formatting.

For the corpus used for TopicGPT, documents retained additional formatting. Titles, post bodies, and comments were labeled and separated by two newline characters (paragraph breaks). Comments were labeled using the format:

Comment from Username: Comment text

When the username was not available, it was replaced with “Deleted author.”

LDA Implementation

LDA was implemented using the gensim Python package. Words were converted to lowercase and tokenized. Words containing only one character or only numbers were removed. Words were also reduced to base forms using WordNetLemmatizer (for example, eating → eat).

Stopwords (common words such as the, or, and if) were removed using the Mallet stopword list, with additional words added following initial runs. A dictionary of words included in the analysis was then created from the resulting tokens, excluding words that appeared in fewer than 20 documents.

The corpus was converted into a bag-of-words representation, where each document becomes a list of counts corresponding to how many times each word in the dictionary appears in that document. Model parameters were chosen based on those described in the reference article.

BERTopic Implementation

BERTopic was implemented using the BERTopic Python library following the official documentation. The default implementation used UMAP dimensionality reduction and clustering. English stopwords from scikit-learn were removed, along with additional custom stopwords that frequently appeared across topics in earlier runs.

An additional model was implemented using k-means clustering, similar to the approach described in another similar article (Park et al., 2018). This allowed the number of clusters to be set to 13 without an outlier cluster.

TopicGPT Implementation

TopicGPT was implemented according to its GitHub documentation and the original paper (Pham et al., 2024). A subset of 600 posts was used following the authors’ recommendations. Prompts and seed topics were tailored using the themes school and medication, which appeared consistently across modeling runs.

Example documents were selected using the representative documents feature of BERTopic, which was also used to identify template examples for the subtopics of social interaction (retail shopping and eye contact.)

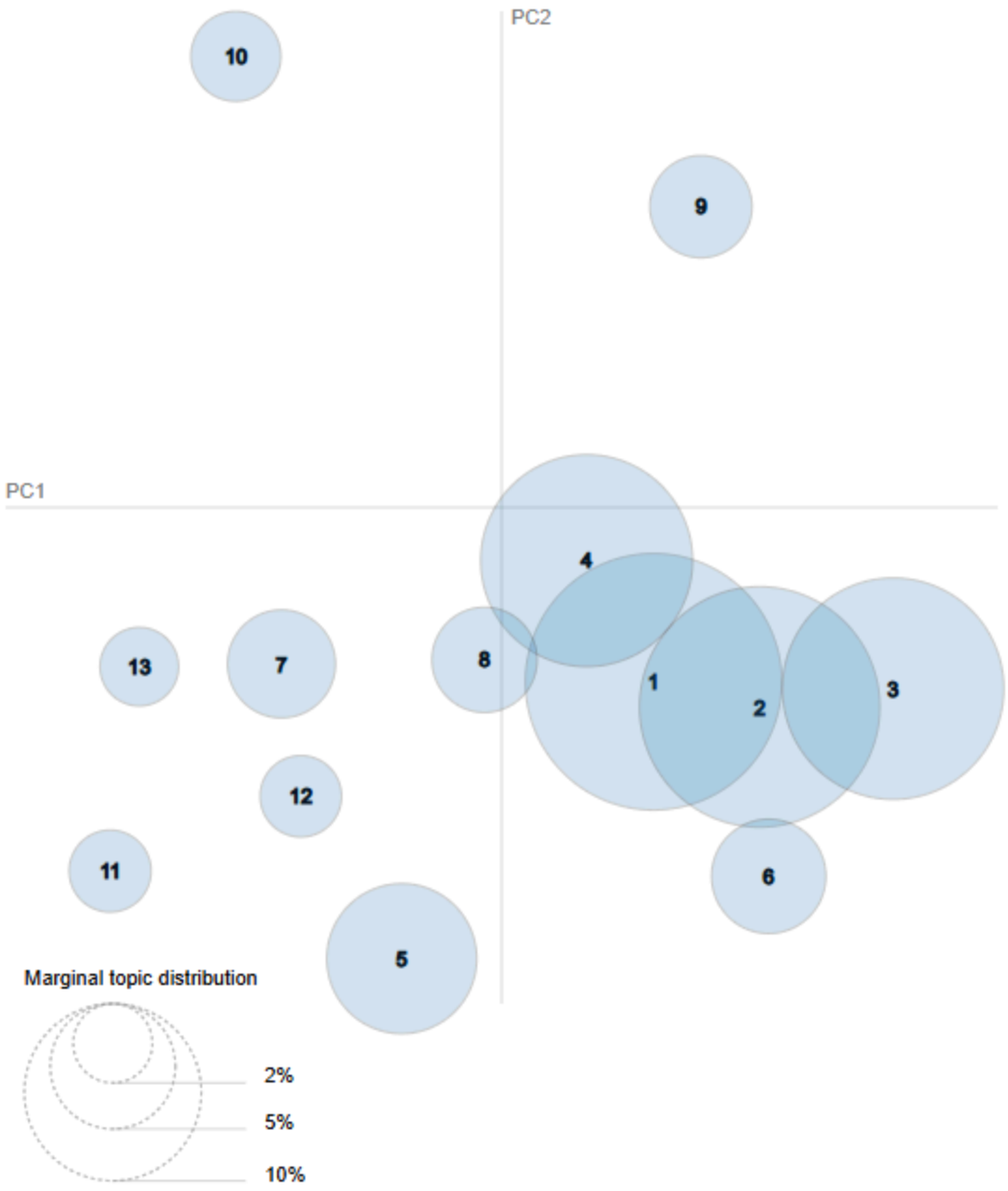
Results

The implemented LDA model resulted in a topic coherence score of 1.8541. Qualitatively, the topics appeared coherent and were widely scattered when visualized using LDAvis. The topics with suggested labels are shown in Figure 1, and the pyLDAvis visualization is provided in Figure 2.

Figure 1: Topics and suggested labels from LDA

Topic	Size (%)	Label	Top Words
1	21	Time and Family	time, day, year, back, felt, started, bad, home, family, week
2	18.4	Life Struggles	life, make, thought, good, person, change, love, care, find, world
3	15.6	SA Symptoms	anxiety, social, anxious, situation, make, fear, time, sa, interaction, thought
4	14.3	Conversations	talk, talking, conversation, make, person, time, award, dont, weird, question
5	7.2	Relationships	friend, group, year, make, game, talk, school, meet, hang, find
6	4.2	Treatment	therapy, anxiety, therapist, medication, doctor, depression, work, med, cbt, mental
7	3.7	School	school, class, teacher, college, presentation, high, grade, year, group, student
8	3.5	Work	job, work, call , interview, phone, working, customer, good, office, money
9	3.3	Physical Symptoms	eye, contact, body, face, make, stop, start, head, smile, gym
10	2.6	Shopping	walk, car, store, walking, food, mask, wear, driving, eat
11	2.1	Dating	girl, guy, woman, date, relationship, dating, drink, girlfriend, crush, boyfriend
12	2.1	Positive Events	good, step, proud, great, music, hope, happy, luck, today, awesome
13	2	Digital Communication	post, online, message, reddit, comment, medium, chat, video, text, posting

Figure 2: LDA topic visualization using pyLDAvis library.



The BERTopic implementation using UMAP clustering produced 90 topics, with 13,217 posts classified as outliers. The first 30 topics are shown in Figure 3, and all topics are stored in the file UMAP topic frequency and words.txt. Using k-means clustering, clusters appeared visually scattered when viewed using the visualize_topics method and appeared qualitatively coherent. These topics are shown in Figures 4 and 5.

Figure 3: Topic 0 through 30 of BERTopic implementation using UMAP clustering.

```
Topic 0(1328): ['job', 'interview', 'jobs', 'work', 'working', 'interviews', 'got', 'good', 'retail', 'customers']
Topic 1(1087): ['friends', 'friend', 'college', 'make', 'school', 'group', 'hang', 'new', 'want', 'lonely']
Topic 2(829): ['family', 'parents', 'mom', 'dad', 'sister', 'mother', 'tell', 'talk', 'want', 'help']
Topic 3(596): ['girl', 'date', 'friend', 'ask', 'said', 'asked', 'crush', 'didn', 'talk', 'good']
Topic 4(554): ['therapist', 'therapy', 'appointment', 'doctor', 'help', 'session', 'therapists', 'cbt', 'psychiatrist', 'sessions']
Topic 5(412): ['medication', 'effects', 'taking', 'meds', 'zoloft', 'dose', 'prescribed', 'doctor', 'depression', 'ssri']
Topic 6(390): ['store', 'order', 'food', 'shopping', 'ordering', 'buy', 'cashier', 'grocery', 'stores', 'shop']
Topic 7(388): ['presentation', 'presentations', 'class', 'speaking', 'speech', 'present', 'presenting', 'public', 'teacher', 'nervous']
Topic 8(357): ['dating', 'date', 'relationship', 'women', 'tinder', 'girls', 'guy', 'girl', 'dates', 'meet']
Topic 9(339): ['therapy', 'exposure', 'help', 'life', 'overcome', 'sa', 'better', 'things', 'fuck', 'make']
Topic 10(329): ['phone', 'calls', 'calling', 'make', 'job', 'answer', 'work', 'making', 'number', 'talking']
Topic 11(303): ['chat', 'discord', 'group', 'pm', 'join', 'play', 'interested', 'dm', 'message', 'omegle']
Topic 12(270): ['party', 'invited', 'parties', 'friend', 'birthday', 'going', 'friends', 'want', 'invite', 'night']
Topic 13(252): ['text', 'texting', 'message', 'messages', 'send', 'reply', 'respond', 'texts', 'email', 'emails']
Topic 14(251): ['quarantine', 'pandemic', 'covid', 'home', 'virus', 'going', 'distancing', 'worse', 'work', 'lockdown']
Topic 15(245): ['eye', 'contact', 'eyes', 'look', 'looking', 'staring', 'stare', 'uncomfortable', 'seconds', 'make']
Topic 16(242): ['alcohol', 'drink', 'drinking', 'drunk', 'sober', 'bar', 'drinks', 'night', 'drugs', 'fun']
Topic 17(192): ['play', 'games', 'game', 'playing', 'chat', 'online', 'discord', 'voice', 'mic', 'gaming']
Topic 18(191): ['media', 'instagram', 'post', 'posting', 'account', 'facebook', 'followers', 'pictures', 'picture', 'accounts']
Topic 19(190): ['book', 'books', 'video', 'alwaysimproving95', 'useful', 'summary', 'animated', 'self', 'awardreportcrosspost', 'read']
Topic 20(189): ['zoom', 'camera', 'classes', 'class', 'online', 'teacher', 'mic', 'video', 'webcam', 'meetings']
Topic 21(183): ['class', 'teacher', 'teachers', 'professor', 'students', 'school', 'classes', 'email', 'answer', 'grade']
Topic 22(181): ['self', 'person', 'esteem', 'hate', 'way', 'care', 'good', 'things', 'say', 'personality']
Topic 23(139): ['room', 'roommates', 'roommate', 'living', 'kitchen', 'apartment', 'live', 'dorm', 'space', 'door']
Topic 24(138): ['proud', 'thank', 'today', 'awesome', 'great', 'good', 'happy', 'congrats', 'did', 'hope']
Topic 25(129): ['gym', 'workout', 'weight', 'going', 'exercise', 'membership', 'doing', 'equipment', 'working', 'workouts']
Topic 26(129): ['suicide', 'life', 'die', 'dreading', 'better', 'lifeline', 'help', 'suicidal', 'want', 'things']
Topic 27(117): ['panic', 'attack', 'attacks', 'heart', 'breathing', 'symptoms', 'physical', 'calm', 'chest', 'breathe']
Topic 28(111): ['hair', 'haircut', 'barber', 'cut', 'hairdresser', 'haircuts', 'salon', 'appointment', 'barbers', 'stylist']
Topic 29(110): ['laugh', 'funny', 'laughing', 'joke', 'jokes', 'humor', 'say', 'make', 'joking', 'fake']
Topic 30(107): ['voice', 'speak', 'hear', 'sound', 'loud', 'speaking', 'louder', 'soft', 'volume', 'quiet']
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Figure 4: Kmeans Clustering with BERTopic (13 topics)

Topic	Size (%)	Label	Top Words
1	12.6	Miscellaneous	way, things, person, want, say, talk, make, self, life, friends
2	12.06	Life Struggles	life, help, things, going, want, better, work, make, way, day
3	11.16	Relationships	friends, school, want, life, make, talk, friend, things, try, college
4	9.48	Dating	girl, friends, friend, want, talk, say, girls, good, person, make
5	9.31	Social Interactions	say, talk, conversation, talking, person, awkward, try, things, make, way
6	7.43	School	class, school, teacher, presentation, going, good, classes, group, did, say
7	6.84	Work	job, work, interview, phone, jobs, good, got, working, day, make
8	6.84	Digital Comm.	post, online, want, media, reddit, chat, friends, play, make, talk
9	5.25	Treatment	therapy, medication, help, therapist, doctor, effects, taking, meds
10	5.09	Eye contact	look, eye, contact, looking, face, wear, gym, eyes, wearing, make
11	4.89	Purchasing Food	eat, food, going, store, room, didn, eating, got, want, way
12	4.56	Family	family, parents, mom, want, life, help, talk, things, school, dad
13	4.49	Social Events	party, going, birthday, friends, want, alcohol, friend, drink, good

Figure 5: Distribution of top words found in 13 topics found using UMAP clustering.

Initial topic generation using TopicGPT resulted in 20 topics that were qualitatively coherent and aligned with themes observed in other modeling approaches (Figure 6 and 7). Generated subtopics resulted in an additional 80 subtopics. After reviewing the topics and subtopics, no duplicate topics were identified, and no additional refinement was required.

Figure 6: Markdown output of topicGPT after initial topic modeling run.

```
[1] School (Count: 165): Discusses experiences, interactions, or situations related to educational settings.
[1] Medication (Count: 57): Mentions psychiatric/medical medications (e.g., beta blockers, ADHD meds, antidepressants, benzodiazepines) including access, effects, and concerns about prescriptior
[1] Work (Count: 139): Anxiety and avoidance behaviors in the workplace (e.g., customer interactions, considering quitting due to symptoms).
[1] Therapy (Count: 107): Finding, choosing, or evaluating a therapist or therapy group; accessing therapy via insurance, provider directories, and personal preferences.
[1] Driving (Count: 9): Discusses anxiety, stress, or experiences related to driving, learning to drive, or taking a driving test.
[1] Family (Count: 93): Discusses social anxiety related to interactions with family members or relatives (e.g., visits, gatherings, talking with family).
[1] Relationships (Count: 116): Discusses romantic relationships, dating, and partner support/strain related to social anxiety.
[1] Body Image (Count: 46): Concerns about physical appearance (e.g., acne, body dissatisfaction) that contribute to anxiety or avoidance in social situations.
[1] Social Skills (Count: 181): Discusses difficulties with conversation, self-disclosure, expressing warmth, or other interpersonal communication behaviors.
[1] Rejection Sensitivity (Count: 140): Discusses fear of rejection, oversensitivity to rejection cues, or perceived/actual rejection in social contexts.
[1] Avoidance (Count: 208): Discusses avoiding social situations or withdrawing due to anxiety, fear, or anticipated negative outcomes.
[1] Substance Use (Count: 40): Discusses using alcohol or other substances to cope/self-medicate, including dependence/addiction.
[1] Bullying (Count: 35): Mentions past experiences of being bullied and how it affects current reactions and boundaries.
[1] Online Social Anxiety (Count: 93): Anxiety specifically related to social interaction in online spaces (e.g., Discord servers, Reddit posting/commenting, Twitch chat, voice communications).
[1] Health Anxiety (Count: 12): Concerns, fear, or preoccupation related to illness, disease, or health outcomes (e.g., COVID-19 infection and lingering symptoms).
[1] Trauma (Count: 26): Experiences of sexual assault/rape and its psychological impact.
[1] Physical Symptoms (Count: 80): Discusses bodily/physiological anxiety symptoms (e.g., shaking, blushing, sweating) that occur in social situations.
[1] People Pleasing (Count: 19): Excessive need to gain approval (e.g., “ass kissing”), prioritizing others’ needs, and difficulty asserting oneself.
[1] Sleep Problems (Count: 8): Discusses insomnia, difficulty falling/staying asleep, or sleep hygiene issues related to anxiety or rumination.
[1] Authority Figures (Count: 17): Anxiety or fear specifically triggered by interactions with authority (e.g., police, security, officials).
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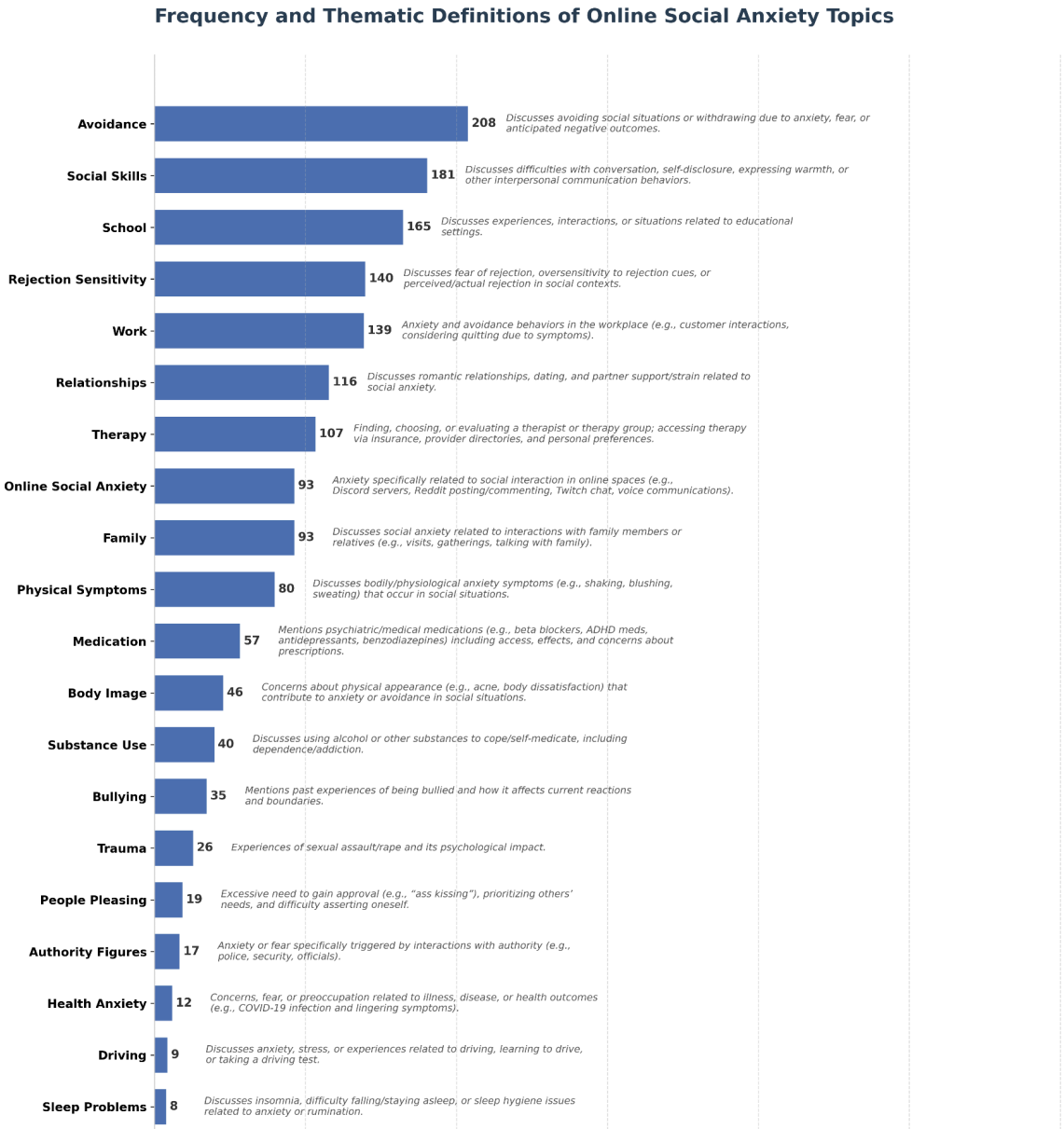
Figure 7: Visualization of topic frequencies from TopicGPT assignments



Discussion

Reproducing the exact model described in the reference paper proved difficult because several important implementation details were not reported, including the stopword list, lemmatization procedure, minimum document frequency thresholds, and model parameters such as decay and offset. In addition, LDA training involves stochastic sampling, meaning that repeated runs with identical parameters can produce different topic assignments. Despite this, viable topic clusters were generated. However, variability between runs was noticeable. For example, medication and therapy were sometimes grouped together and sometimes separated into distinct topics. Similarly, clusters related to school and work consistently appeared but varied in size across runs.

Using a relatively high number of topics in BERTopic allowed for the identification of more specific subthemes within the dataset. Although many documents were excluded as outliers when using UMAP clustering, this outcome appeared reasonable given that the largest topic identified in the reference paper was labeled “miscellaneous.” Several potentially interesting subtopics emerged, including discussions of social status (topic 35), embarrassment (topic 41), and suicidal ideation (topic 26). BERTopic was also the most intuitive model to analyze. In addition to listing representative words, the implementation provides three highly representative documents for each topic along with convenient access to all documents within a topic. Visualization tools were also easier to use than those in LDAvis, and BERTopic ran significantly faster than LDA, especially when experimenting with larger numbers of topics.

TopicGPT produced a larger number of topics than the other methods, some of which appeared overly specific, such as topics describing nervousness around authority figures. Future work could potentially reduce the number of topics through additional refinement steps, adjustments to prompts and seed topics, or the use of larger document samples. Despite this, TopicGPT demonstrated a notable strength in identifying structured subtopics within broader themes. In particular, the model effectively categorized different physical symptoms of social anxiety when generating subtopics, as shown in Figure 8. Because TopicGPT relies on a language model to interpret documents individually, it can produce descriptive topic labels and hierarchical structures, although this approach is more computationally intensive and sensitive to prompt design.

Figure 8: Subset (Physical Symptoms) of second markdown output of TopicGPT which generated subtopics.

```
[1] Physical Symptoms (Count: 80): Discusses bodily/physiological anxiety symptoms (e.g., shaking, blushing, sweating) that occur in social situations.
  [2] Trembling & Shaking (Count: 6): Involuntary shaking/tremors (head, hands, legs) triggered by anxiety or social pressure.
  [2] Cardiovascular Symptoms (Count: 5): Rapid heartbeat/palpitations, pounding chest, feeling faint/blackout during anxious moments.
  [2] Breathing & Chest Tightness (Count: 3): Shortness of breath, “can’t breathe,” heavy/pressure feeling in chest/sternum.
  [2] Gastrointestinal Symptoms (Count: 6): Nausea/vomiting urge, stomach drop/butterflies, diarrhea, appetite disruption linked to anxiety.
  [2] Sweating & Hot Flashes (Count: 5): Sweaty palms/feet, flushing/feeling hot, temperature surges during social stress.
  [2] Facial Tension & Jaw/Throat Tightness (Count: 3): Muscle stiffness in face/jaw/throat that affects speaking/smiling and increases self-consciousne
  [2] Blushing & Flushing (Count: 3): Visible blushing/face and ear heat, often worsened when noticed or called out.
  [2] Eye Watering & Sensitivity (Count: 2): Teary eyes/pressure, eye sensitivity, fear of appearing like crying in public.
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Overall, each modeling approach demonstrated different strengths depending on the intended research goal. Both LDA and BERTopic were effective for identifying broader themes within the dataset and would likely be appropriate for studies seeking high-level topic summaries across large corpora. BERTopic was significantly faster to run and provided more intuitive tools for exploring topics, suh as representative documents. Initial runs of TopicGPT produced less desirable results, but the method may still be effective if additional effort is put into guiding topic generation through prompts and cseed topics. This approach is also more likely to incur computational costs but may yield useful results with further refinement.

For exploratory analyses aimed at identifying more specific subthemes, BERTopic appeared to be the most practical option, particularly when computational resources or time are limited, followed by TopicGPT. Attempts to run LDA with higher numbers of topics were abandoned due to the time required for training. A notable strength of TopicGPT was its high level of interpretability, which may be particularly valuable when examining ambiguous topics or documents that would otherwise be treated as miscellaneous or outliers.

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